Appendix 1

LESSON PLAN

Unit/Lesson Title: Unit 2 Statistics and Measurement **Lesson 12: Surface area of right prisms and cylinders**	**Lesson duration 55 min**	**Stage 5 Year 9**	**Class/Group Mathematics (09MAT_01)**
Rationale This lesson develops students' understanding of surface area by decomposing 3D objects into familiar 2D shapes. Through right prisms and cylinders, students build geometric reasoning, connect nets to area formulas and apply mathematical thinking to practical contexts involving wrapping, covering and design	**Syllabus Outcomes** AC9M9M01	**Syllabus Content** Solve problems involving the volume and surface area of right prisms and cylinders using appropriate units	

Prior Knowledge	**Risk Assessment**	**Resources**
Perimeter, area of 2D shapes, circle circumference and area formulas, decomposition of composite shapes from earlier topics in this unit	N/A	Real-life 3D object (soda can, box), calculators, student notebooks, textbook, whiteboard and markers, projector screen

Learning Indicators*/specific content to be taught*	**Assessment & Recording Strategies**	**Time Guide**	**Learning Experience** *(what the teacher and learners will be doing)*	**Teaching Strategies**	**Class Organisation**
Identify the surfaces contributing to surface area of prisms/cylinders Use nets and decomposition to derive surface area formulas Calculate surface area of right prisms and cylinders	Formative assessment: Teacher observation during class discussion and individual work. Identify common misconception to address during this lesson or future lessons.	5 min 10 min	*Introduction (Engagement/Motivation/Purpose)* Teacher displays familiar objects (milk box, soda can) and asks students how manufacturers determine the amount of material needed to wrap or cover objects. Teacher nominates a number of students to offer their suggestions. In the class discussion, teacher guide students to distinguish between surface area and volume and introduce the idea of decomposing 3D objects into simpler 2D surfaces. *Body (Exploration/Transformation/Presentation/Practice)* I do: explicit teaching/modelling Teacher models how surface area of a rectangular prism can be found by identifying repeated faces and decomposing the solid into rectangles	Teacher questioning Real-world contextualisation Class discussion Explicit teaching Gradual lease model (I do, we do, you do) Teacher modelling	

	Formative assessment: Exit ticket informs teacher of level of engagement and misconceptions to address in the future	10 min 20 min 10 min	Teacher introduces nets as a visual representation of surfaces and derives the prism surface area formula $A = 2lw + 2wh + 2lh$ collaboratively with students. Demonstration done using marker on whiteboard + projection of presentation on screen using an iPad. Teacher transitions to cylinder by discussing how curved surface can be unrolled into a rectangle whose length equals the circumference of the base. This is followed by deriving cylinder surface area formula visually and model one worked example. We do: guided practice Student work within table groups to solve a prism and cylinder surface area problem. Teacher chooses moments to pause the activity and use questioning to guide students toward the solution. You do: independent practice Student works individually on a number of practice problems on the textbook involving prisms and cylinders which gradually builds proficiency and increase in difficulty. Brief interchanges between table groups are permitted, provided it does not affect other students' work. Teacher walks around to monitor student progress and assess their understanding. Providing feedback and support students as needed. *Conclusion (Presentation/Reflection/Review)* Class review key ideas involving surface area, nets and techniques in computing surface area. Student completes a brief online exit ticket	Worked examples Guided practice / scaffolding Collaborative learning Peer discussion Formative assessment	
Transition to next lesson					

Associate Teacher's Signature…… **Date**……
NB. ***This must be signed before the lesson is taught***